

Torque Rebalance — math and algorithm

b601_teleop / b601/assist.py :: TorqueRebalance — designed 2026-08-26

Goal. Isotropic drag feel: a hand force at the gripper should translate the EE the same way in every direction, instead of escaping as cheap wrist rotation.

Notation

symbol	meaning
$q, \dot{q} \in \mathbb{R}^6$	joint positions and velocities (joint1...joint6)
$J \in \mathbb{R}^{3 \times 6}$	position Jacobian of <code>gripper_end</code> , world frame: $v_{ee} = J\dot{q}$
J_i	column i : direction & lever with which joint i moves the EE
$B = J_{[:, 1:3]}$	upstream columns (base/shoulder, joints 1–3)
$W = J_{[:, 4:6]}$	downstream columns (wrist, joints 4–6)
$\odot$	elementwise product
$\text{lp}_T(x)$	first-order low-pass: $x \leftarrow x + \frac{dt}{T+dt}(x_{raw} - x)$

Physical background — why the wrist "does everything"

A hand force F at the EE loads **every** joint at once through the transpose map (virtual work):

$$\tau_i = J_i^\top F$$

Joint i yields only when $|\tau_i|$ exceeds its own breakaway friction b_i . The chain yields at its weakest link: the wrist crosses its threshold at a lower push force (lower friction; for radial pushes also a bigger lever), the gripper then follows the hand, and the contact force plateaus **below** the base joints' threshold. Additionally, while sliding, each joint's speed per newton is

$$\dot{q}_i = \frac{J_i^\top F - b_i}{k_{d,i}}$$

and the wrist's slope (lever/ k_d) is ~20× the shoulder's, so the wrist supplies the hand's motion first and the force never climbs.

Step 1 — measure the external torque on the downstream joints

External torque is observable only through **backdrive**: a stuck joint hides it inside its static friction band, and in free-float drag ($k_p = 0$) the motor does not resist a push, so torque feedback cannot see it either.

While a joint moves, the hand torque on it is exactly what sustains the motion:

$$\tau_{dn} = f_{dn} \odot \tanh\left(\frac{\dot{q}_{dn}}{v_0}\right) + k_{dn} \odot \dot{q}_{dn}, \quad \tau_{dn} \leftarrow \text{lp}_{\tau_f}(\tau_{dn})$$

Zero at rest by construction. $f_{dn} = [0.30, 0.21, 0.21]$ N·m (Seed friction sweep, j4/j5/j6); k_{dn} = the `kd_drag` actually commanded on the wrist (default 0.05).

Step 2 — infer the external torque on the upstream joints (Jacobian)

The upstream joints are the stuck ones, so their external torque cannot be measured; it is reconstructed. The hand force explaining the downstream torques is the damped least-squares solution of $W^\top F = \tau_{dn}$:

$$A = W^\top, \quad \hat{F} = (A^\top A + \lambda^2 I)^{-1} A^\top \tau_{dn}, \quad \tau_{up} = B^\top \hat{F}$$

Observability caveat: only the force component in the span of the *backdriving* joints' lever rows is recoverable (minimum-norm elsewhere). One yielding wrist joint $\Rightarrow$ 1-D information — e.g. a pure-outward push and an out-and-up push give identical j4 torque and are indistinguishable. This is why step 4 does **not** take its direction from $\hat{F}$.

Step 3 — compare the halves $\rightarrow$ transfer gate w

Two rejected comparisons: *raw torque vs raw torque* (a joint yields against its **own** breakaway; laterally joint1 receives more raw torque than joint5 and still loses) and *threshold-normalized loading vs loading* (the step-2 ambiguity inflates the inferred upstream loading, and a loaded-but-stuck joint is not helping anyway — loading is not carriage). The robust comparison is what each half is actually **doing**:

$$r_{down} = \max_i \frac{|\tau_{dn,i}|}{b_{dn,i}}, \quad c_{up} = \frac{\|B\dot{q}_{up}\|}{\|B\dot{q}_{up}\| + \|W\dot{q}_{dn}\| + \varepsilon}$$

$$w = \text{clip}(r_{down}, 0, 1) \cdot (1 - c_{up}) \in [0, 1]$$

$w \approx 1$: wrist carries the push alone $\rightarrow$ transfer fully. $w \rightarrow 0$: upstream has taken over the motion $\rightarrow$ stop pushing it. $w = 0$: no hand $\rightarrow$ no output.

Step 4 — transfer ($J^\top$ size, J^+ direction)

The two Jacobian maps answer different questions and can **disagree in sign** in skewed geometry (folded-wrist radial push: a force along the motion loads j2 negative, but j2 must rotate positive to carry that motion):

- $J^\top F$ — torque the force **exerts** on a joint (statics)
- $J^+ u$ — rotation the joint needs to **carry** motion u (kinematics)

So: per-joint direction pattern from the motion-carry map, magnitude from the sensed force.

$$u_m = \frac{W \dot{q}_{dn}}{\|W \dot{q}_{dn}\|} \quad (\text{EE direction the wrist is carrying})$$

$$p = B^\top (BB^\top + \lambda^2 I)^{-1} u_m \quad (\text{upstream rates that carry } u_m)$$

$$B^\top F_{eq} = p \text{ (DLS)}, \quad s = \gamma \frac{\|\hat{F}\|}{\|F_{eq}\|} \quad (\text{pattern's EE force} = \gamma \times \text{sensed force})$$

$$\tau_{tx} = \text{clip}\left(\text{lp}_{\tau_f}(w \cdot s \cdot p), \pm \tau_{max}\right) \quad \text{per joint 1-3}$$

Injected into the drag torque: $\tau_{cmd} = g(q) + f_{comp} \odot \tanh(\dot{q}/0.08) + \tau_{tx}$.

Safety properties (all tested in `tests/test_assist.py`)

- **Zero at rest:** no backdrive $\Rightarrow \tau_{dn} = 0, w = 0, u_m$ undefined $\Rightarrow$ no output.
- **Bounded:** EE-equivalent force $\leq \gamma \times$ sensed hand force; per-joint τ_{max} cap; controller clamp and `vel_abort` freeze as backstops.
- **Self-fading:** gate closes as the upstream half takes over; the u_m source vanishes when the wrist stops slipping.
- **No wrong-way pressing:** every driven joint aids the observed motion (J^+ sign).
- No rate targets, no integrator, no added damping.

Pseudocode (per control cycle, 100 Hz, DRAG phase only)

```

state: tau_dn = 0 (R^3), tau_tx = 0 (R^3)          # low-pass filter states

update(q, qd, dt):
    Jv = jacobian(q)[:3]                          # 3x6
    B, W = Jv[:, :3], Jv[:, 3:]

    # step 1: measure downstream external torque from backdrive
    raw    = fric_dn * tanh(qd[3:] / v0) + kd_dn * qd[3:]
    tau_dn = lowpass(tau_dn, raw, tau_f, dt)

    # step 2: infer hand force and upstream received torque
    A      = W.T
    F_hat  = solve(A.T A + lam2*I3, A.T tau_dn)
    tau_up = B.T @ F_hat                          # displayed; sizing input

    # step 3: compare halves -> gate
    r_down  = max(|tau_dn| / brk_dn)
    carry_up = |B qd[:3]| / (|B qd[:3]| + |W qd[3:]| + eps)
    w       = clip(r_down, 0, 1) * (1 - carry_up)

    # step 4: transfer (J^T size, J+ direction)
    v_dn = W @ qd[3:]
    if |v_dn| < 1e-4: raw_tx = 0
    else:
        u_m    = v_dn / |v_dn|
        p      = B.T @ solve(B B.T + lam2*I3, u_m)
        F_eq   = solve(B B.T-style DLS of B.T F = p)
        s      = gain * |F_hat| / |F_eq|
        raw_tx = w * s * p
    tau_tx = clip(lowpass(tau_tx, raw_tx, tau_f, dt), -tau_max, tau_max)

    return [tau_tx, 0, 0, 0]                      # torque on joints 1..3 only

```

Parameters (defaults)

param	default	meaning
γ (gain)	2.0	force multiplication wrist $\rightarrow$ base (<code>--assist-gain</code>)
τ_{max}	1.2 N·m	cap per upstream joint (the <code>--assist TAU</code> argument)
f_{dn}, b_{dn}	[0.30, 0.21, 0.21] N·m	wrist Coulomb friction / breakaway (Seed sweep)

param	default	meaning
k_{dn}	[0.05, 0.05, 0.05]	wrist <code>kd_drag</code> actually commanded
v_0	0.08 rad/s	tanh knee (above finite-difference velocity noise)
τ_f	0.1 s	low-pass time constant (measurement and output)
λ	0.05 m	damping of every least-squares solve

Results (sim, identical 2 N pushes at the folded-wrist pose $q = [0, 1.08, 1.22, -1, 0, 0]$)

direction	plain	rebalanced
lateral	0.263 m	0.269 m
radial	0.085 m	0.148 m (+74%, the joint4 direction)
vertical	0.233 m	0.233 m

Direction spread $3.1\times \rightarrow 1.8\times$. High-stiction j1 (0.8 N·m) recruited with peak 0.49 N·m transfer, zero torque sign flips, stops with the hand.

Run

```
python scripts/gravity_drag.py drag --assist 1.2 [--assist-gain 2]
```

Status line: `tau_dn <j4 j5 j6> -> tau_up <j1 j2 j3> w <gate> tx <j1 j2 j3>`